

Network Analysis of the Bitcoin OTC Trust Weighted Signed Network Dataset

1. Introduction

1.1 About the Dataset

The Bitcoin OTC Trust Weighted Signed Network dataset represents a **who-trusts-whom** network of individuals engaging in Bitcoin trading on the Bitcoin OTC platform. Given Bitcoin's decentralized and pseudonymous nature, trust plays a crucial role in ensuring secure and reliable transactions. The dataset records **trust ratings** assigned by users to others, helping to establish a reputation system that mitigates fraudulent activities.

- **Trust ratings range from -10 (total distrust) to +10 (total trust)** in steps of 1.
- Understanding trust dynamics in such a network provides valuable insights into **user behavior, fraud detection, and risk assessment**.

1.2 Key Terms and Definitions

To better understand the analysis, it is essential to define key concepts:

- **Network:** A system of interconnected entities (nodes) and their relationships (edges).
- **Node:** A point in the network representing an individual or entity.
- **Edge:** A connection between two nodes that represents an interaction.
- **Directed Network:** A network where relationships between nodes have a direction.
- **Undirected Network:** A network where connections have no direction, meaning interactions are mutual.
- **Weighted Network:** A network where edges carry numerical values (in this case, trust ratings).
- **Signed Network:** A network where edges can have positive (trust) or negative (distrust) values.
- **Betweenness Centrality:** A measure of how often a node appears on the shortest paths between other nodes, indicating its importance as a bridge.
- **Degree Centrality:** A measure of a node's connectivity by counting the number of edges.
- **PageRank:** A ranking algorithm that assigns influence scores based on incoming connections.
- **Community Detection (Louvain Algorithm):** A method used to identify clusters or groups of nodes with strong internal connections.
- **Density:** The ratio of actual connections to possible connections within the network.
- **Clustering Coefficient:** A measure of the tendency of nodes to form tightly-knit groups.

1.3 Justification for Graph Type Used

- **Directed Graph:** Trust relationships are one-way, meaning that user A may trust user B, but user B may not necessarily trust user A.
- **Undirected Graph for Community Detection:** Helps in identifying groups with strong trust connections without considering directionality.

1.4 Dataset Structure and Its Significance

The dataset consists of:

- **Source (Node) - Trustor:** The user assigning the rating.
- **Target (Node) - Trustee:** The user receiving the rating.
- **Rating (Edge Weight):** The score assigned by the trustor to the trustee.
- **Timestamp:** The date and time when the rating was recorded.

Understanding these elements is crucial for analysing how trust relationships evolve and influence network dynamics.

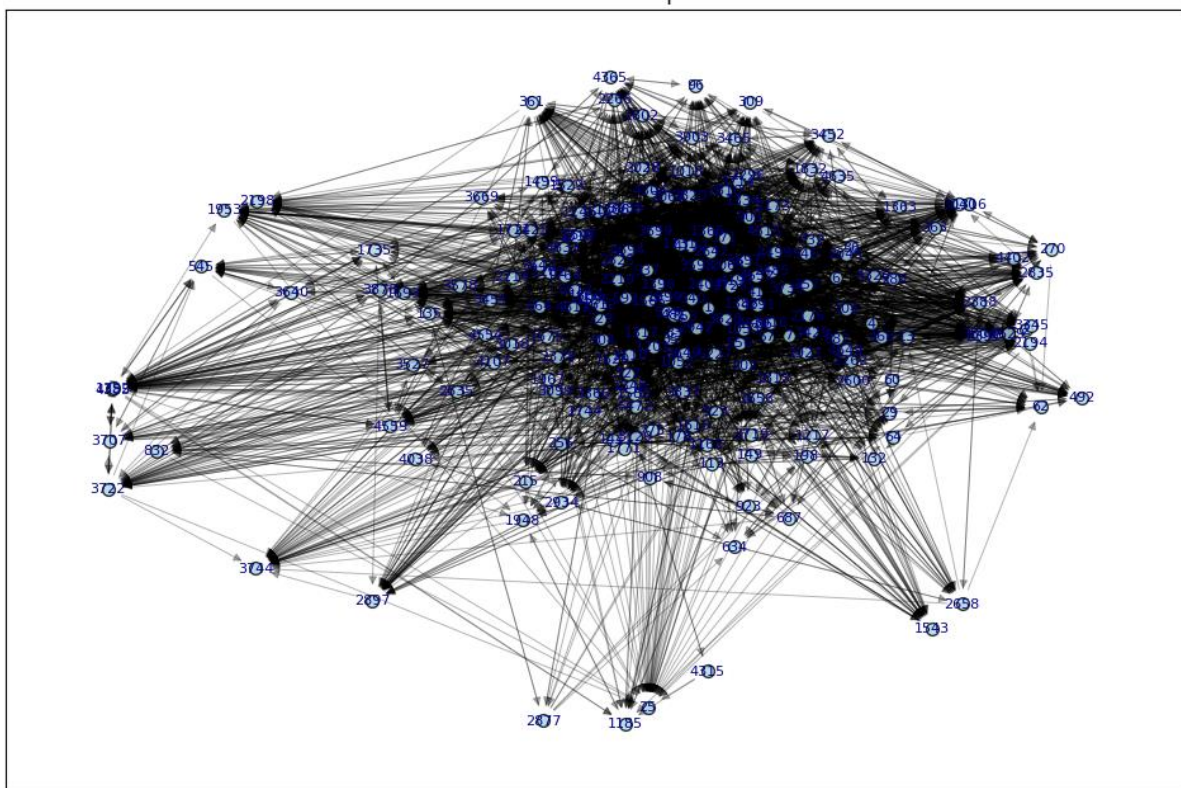
2. Network Properties and Metrics

2.1 Overall Network Characteristics

- **Total Nodes (Users):** 5,881
- **Total Edges (Trust Relationships):** 35,592
- **Network Density:** 0.0010 (Sparse network, meaning trust relationships are selective)
- **Average Clustering Coefficient:** 0.178 (Moderate connectivity forming trust communities)

These metrics indicate that the Bitcoin OTC network is **highly sparse**, suggesting that trust relationships are selective rather than widespread. Users tend to engage in transactions with a limited number of trusted counterparts, reinforcing the importance of community structures.

Bitcoin Trust Network - Top 200 Users



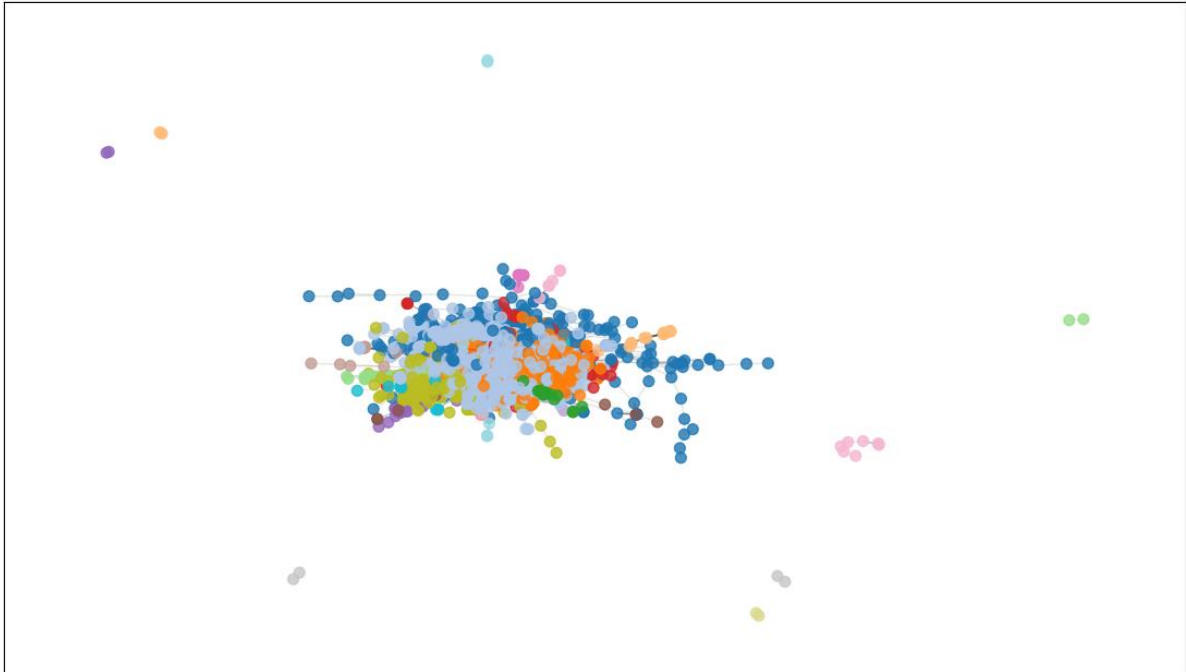
2.2 Key Observations from Network Structure

- **Low Density:** Indicates that trust is not distributed evenly but is concentrated among specific users.
- **Moderate Clustering Coefficient:** Suggests that some groups form interconnected clusters, reinforcing local trust.
- **Highly Connected Influencers:** A few key users dominate the network, shaping overall trust patterns.

3. Research Questions and Insights

3.1 How is Trust Distributed Among Users in the Bitcoin OTC Network?

Using Louvain community detection:



- **Trust is localized:** Users prefer to interact within their communities rather than across the entire network.
- **Key intermediaries bridge groups:** Some users act as connectors between different communities, facilitating trust.
- **Peripheral users exist:** Some users remain isolated, possibly due to being new or having low trust ratings.
- **Influential users shape credibility:** A handful of users play a crucial role in maintaining trust within their communities.

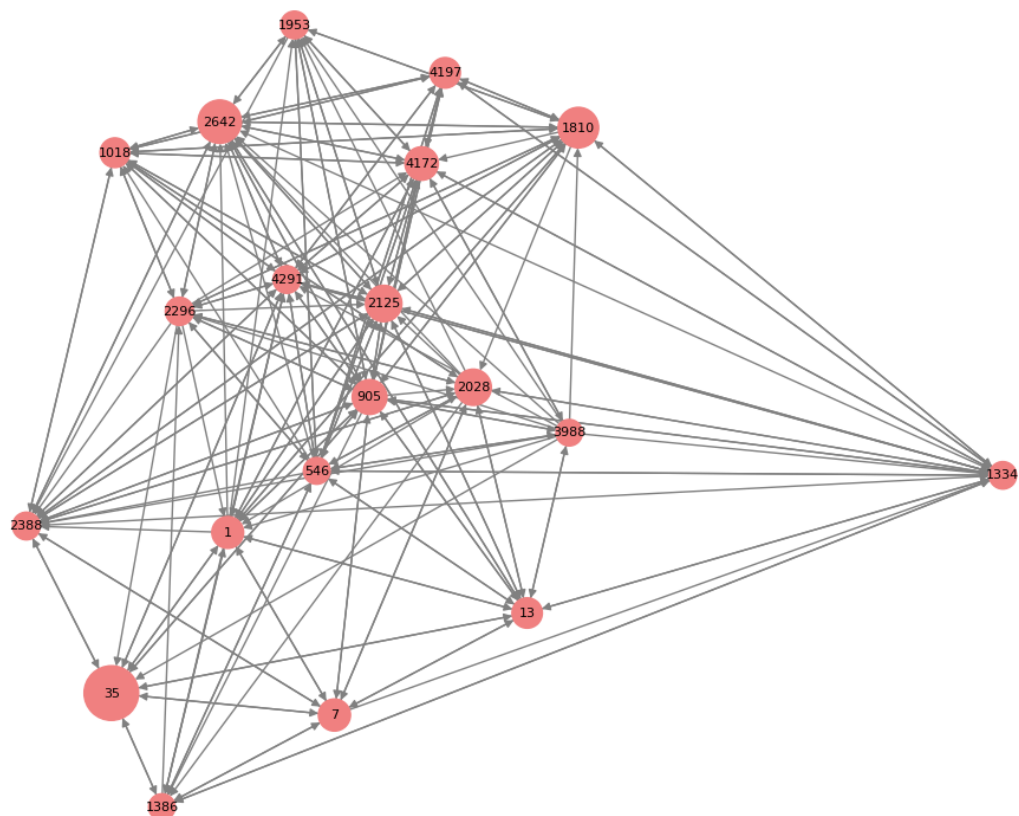
This analysis suggests that trust in the Bitcoin OTC network is **not evenly distributed** but instead concentrated in tightly-knit groups, where a few highly trusted individuals dominate interactions.

3.2 Who Are the Most Connected, Influential and key intermediate Users in the Trust Network?

Using **degree centrality**, **PageRank**, and **betweenness centrality**, the most influential, connected and key intermediate users were identified:

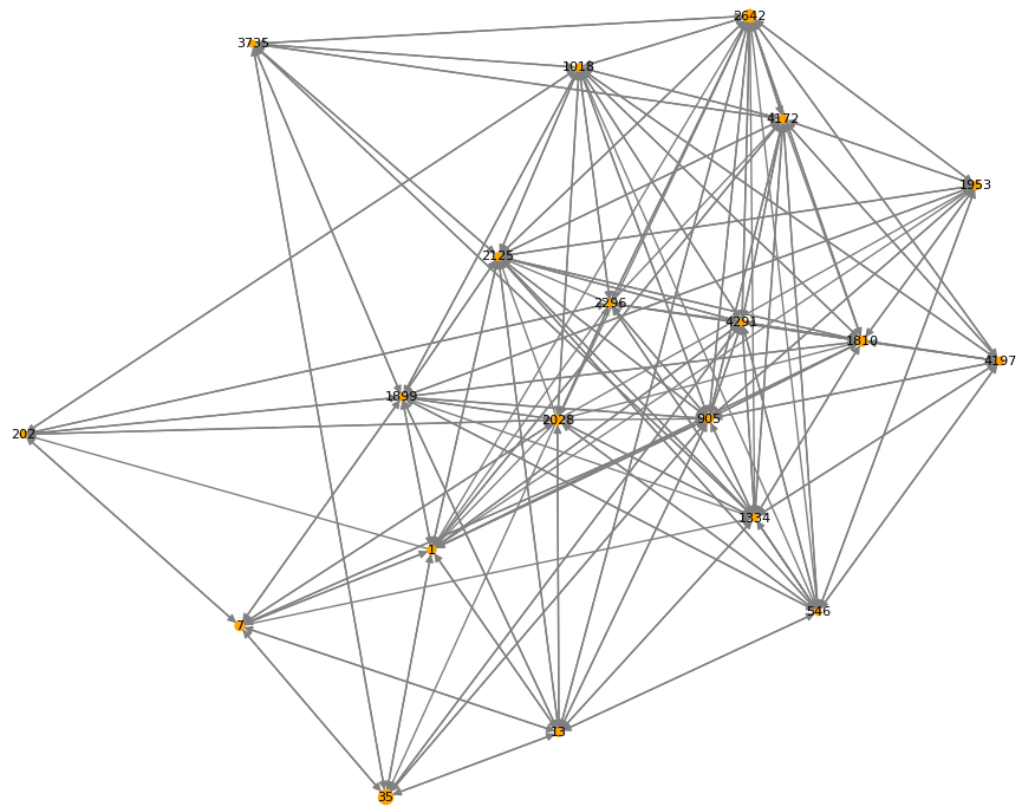
- **Degree Centrality (Most Connected Users):** User 35 leads, followed by 2642, 1810, 2125, and 2028. These users have the highest number of direct connections, making them central hubs

Top 20 Nodes by Degree Centrality in the Bitcoin Trust Network

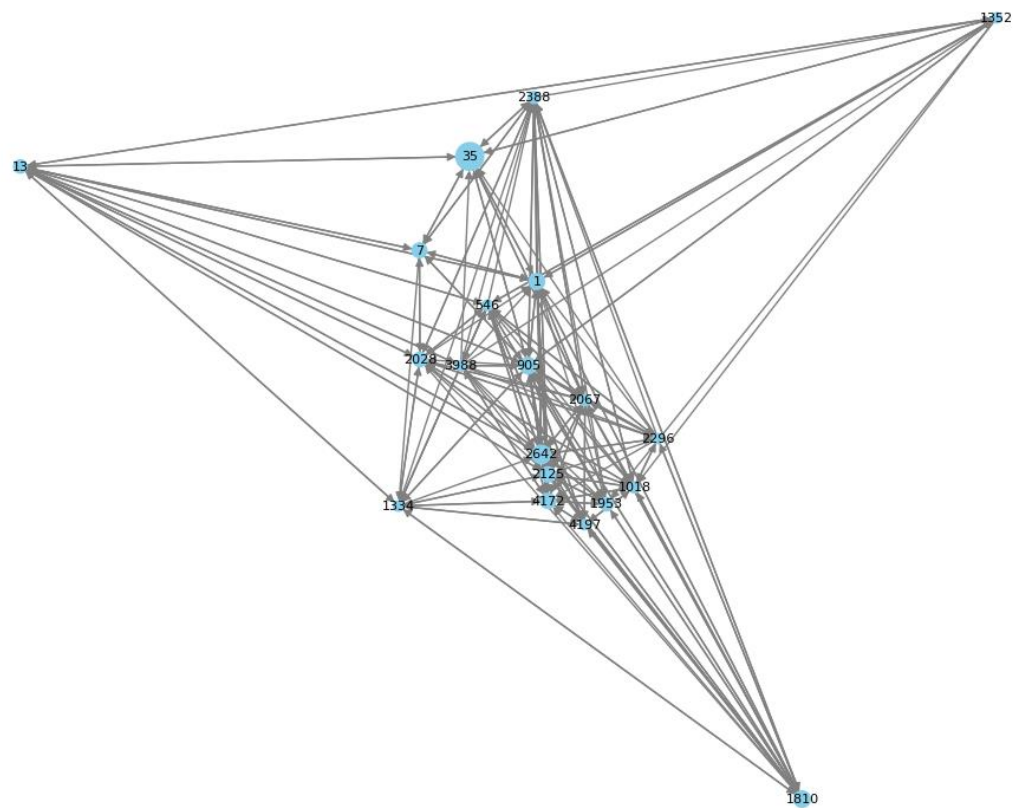


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- PageRank (Most Trusted Users):** User 35 ranks highest, followed by 2642, 1810, and 2028. This ranking highlights users trusted by other influential individuals
 Top 20 Influential Users in the Bitcoin Trust Network (PageRank)



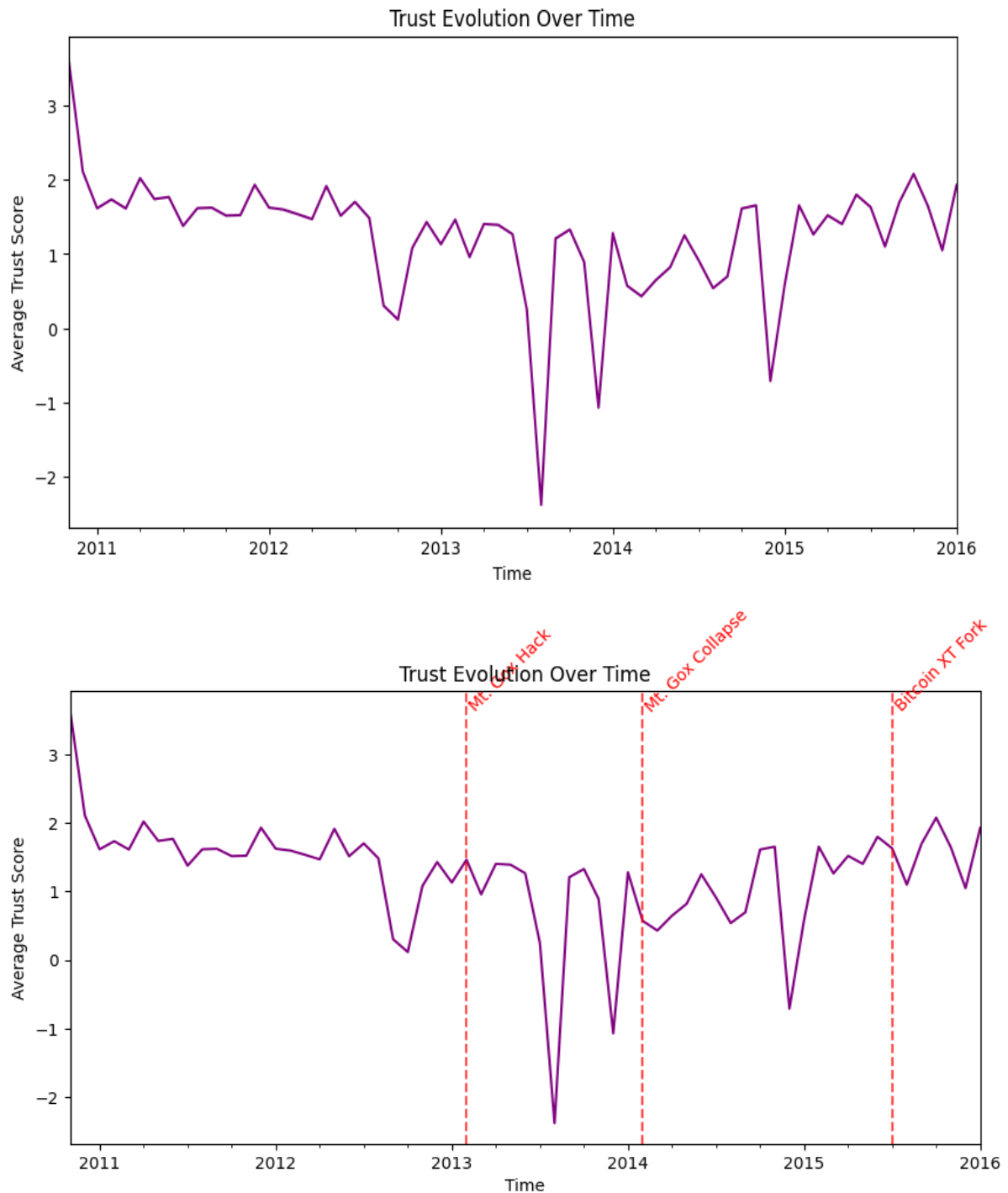
- **Betweenness Centrality (Key Intermediaries):** User 35 is the most critical bridge between different clusters, followed by 2642, 1810, 905, and 1.
Top 20 Intermediaries in the Bitcoin Trust Network



Interpretation:

- The presence of **centralized trust hubs** makes the network vulnerable – if these users leave or engage in fraudulent activities, trust in the network could collapse.

3.3 How Does Trust Change Over Time in the Network?



Historical trust fluctuations reveal significant declines during major Bitcoin events:

Analysis: Examined trust fluctuations during major Bitcoin events:

- **Mt. Gox Hack (Feb 2013):** Caused a notable drop in trust.
- **Mt. Gox Collapse (Feb 2014):** Led to a sharp decline in trust across the network.
- **Bitcoin XT Fork (July 2015):** Introduced instability, negatively impacting trust levels.

Key Insight:

- External shocks significantly impact user trust, leading to fragmentation and reduced interactions.

3.4 How Do Users Regain Trust After a Decline?

```
#How do users regain trust after a decline?
Initial_trust = df[df.index < "2013-01-01"].groupby("source")["rating"].mean()
final_trust = df[df.index > "2015-01-01"].groupby("source")["rating"].mean()
trust_recovery = pd.DataFrame({"Initial": Initial_trust, "Final": final_trust}).dropna()
trust_recovery["Change"] = trust_recovery["Final"] - trust_recovery["Initial"]

print("\nTop 10 Users who Gained the Most Trust Over Time:")
print(trust_recovery.sort_values("Change", ascending=False).head(10))
```

source	Initial	Final	Change
481	-5.250000	2.666667	7.916667
1819	2.300000	9.000000	6.700000
2067	-3.438716	2.142857	5.601573
2	-3.047019	8.000000	4.952381
2073	1.333333	5.500000	4.166667
1810	-1.557895	1.625000	3.182895
2552	-0.142857	2.666667	2.809524
1639	-0.142857	2.600000	2.742857
1565	1.648649	4.375000	2.726351
2731	1.600000	3.666667	2.066667

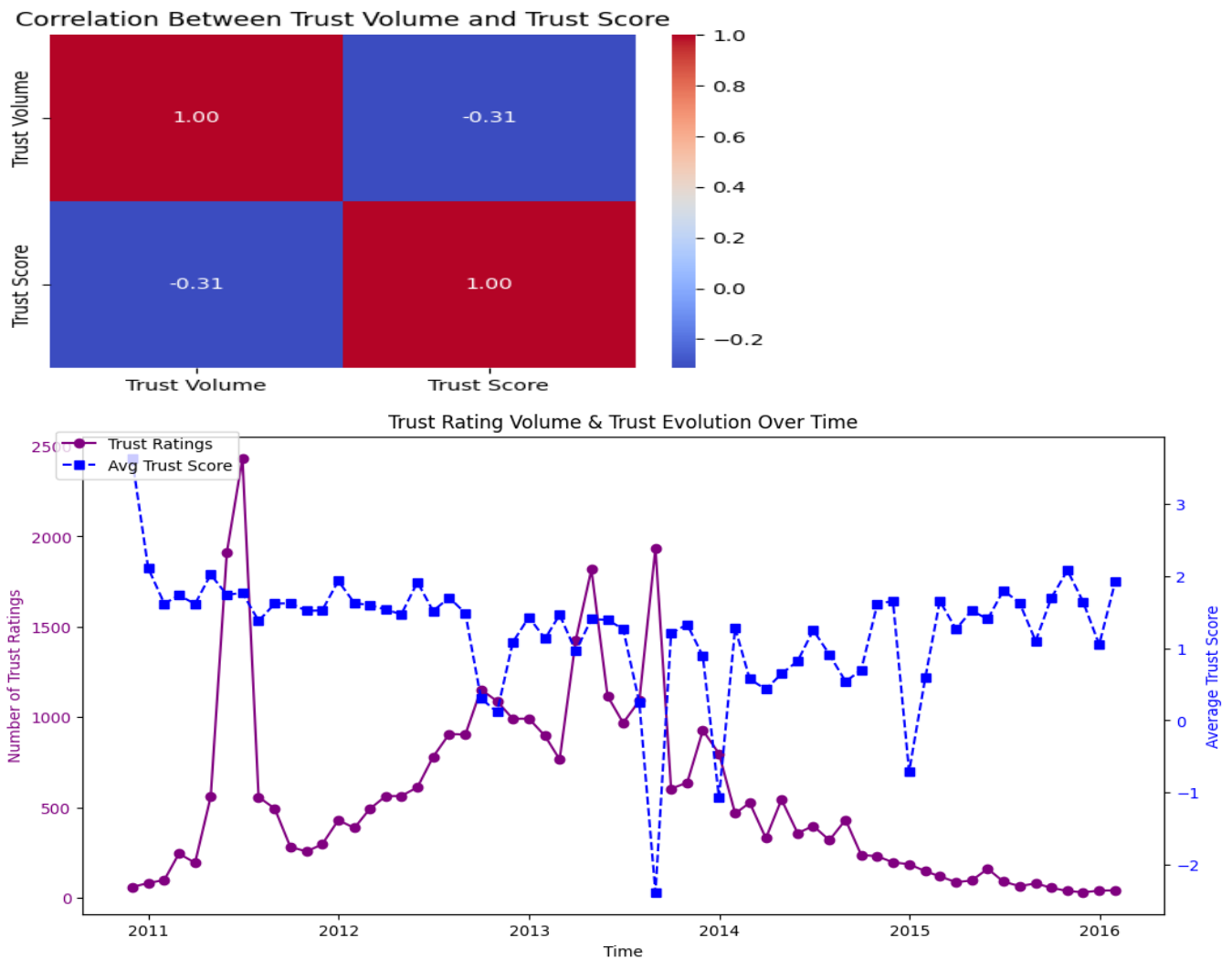
Findings:

- Some users successfully rebuild their reputations:
 - User 481 improved from **-5.25 to 2.67** (+7.92 shift).
 - User 1819 surged from **2.30 to 9.00** (+6.70 increase).
- Trust recovery is often linked to positive **transactions** and endorsements from already trusted users.

Key Insight:

- Trust recovery is possible but challenging – users need consistent positive interactions to rebuild reputation.

3.5 Does More Trust Ratings Mean Higher Trust?



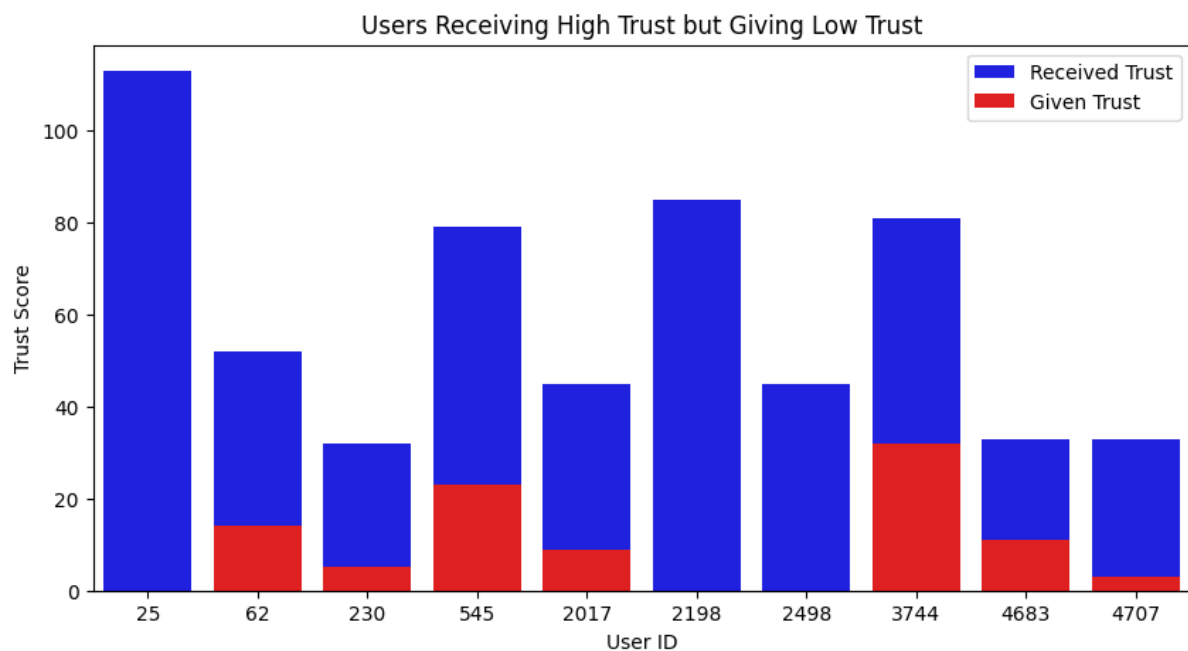
Findings:

- Weak negative correlation (-0.31) between the number of trust ratings and overall trust score.
- Simply receiving more ratings does not necessarily improve trustworthiness.

Key Insight:

- Users who receive many ratings are not necessarily more trusted – the quality of interactions matters more than quantity.

3.6 Are There Users Who Receive High Trust but Give Low Trust?



Findings:

- Some users benefit from high trust but rarely endorse others.
- This behavior suggests either cautious engagement or strategic trust hoarding.

Interpretation:

- These users could be influential decision-makers who carefully select whom they trust.
- Alternatively, they may be exploiting the network's trust system for personal gain.

4. Conclusion and Implications

Key Takeaways:

- Trust is highly centralized, with a few key influencers shaping network credibility.
- Trust levels fluctuate significantly due to external events, such as the Mt. Gox collapse.
- Users can recover trust over time, but it requires consistent positive engagement.
- The network is vulnerable to manipulation due to selective trust dynamics.

Real-World Applications:

- **Fraud Detection:** Identifying trust hoarders or users manipulating ratings can help detect fraudulent activity.
- **Risk Assessment:** Decentralized platforms can implement automated trust score monitoring to detect anomalies.
- **Reputation Systems:** Platforms like Bitcoin OTC could reward users who distribute trust fairly to promote a healthier network.

Next Steps:

- Further analysis into automated fraud detection based on trust rating anomalies.
- Predictive modeling to assess which users are at risk of trust decline.

Final Thoughts:

By analyzing trust in decentralized networks, we can create safer, more reliable platforms for digital transactions. Understanding these dynamics helps in building resilient systems that can withstand shocks and fraudulent behavior.